

NUR-SULTAN, Kazakhstan

WMO: 351880

Lat: 51.1703N

Lon: 71.3942E

Elev: 350

StdP: 97.19

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.0	(c) -30.1	(d) -36.8	(e) 0.1	(f) -33.0	(g) -33.3	(h) 0.2	(i) -29.9	(j) 14.7	(k) -5.8	(l) 13.1	(m) -6.1	(n) 2.9	(o) 60	(p) 0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.1	(c) 32.3	(d) 17.8	(e) 30.5	(f) 17.3	(g) 28.7	(h) 16.9	(i) 19.6	(j) 27.6	(k) 18.8	(l) 26.6	(m) 18.1	(n) 25.7	(o) 3.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.8	21.6	16.2	12.0	21.1	15.2	11.3	20.8	57.4	27.2	55.0	26.7	52.5	25.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.3	(b) 10.5	(c) 9.2	(e) -36.0	(f) 35.9	(g) 3.1	(h) 2.4	(i) -38.2	(j) 37.6	(k) -40.0	(l) 39.0	(m) -41.8	(n) 40.3	(o) -44.0	(p) 42.1		
(a) 12.3	(b) 10.5	(c) 9.2	(e) -36.2	(f) 21.1	(g) 3.1	(h) 1.0	(i) -38.4	(j) 21.8	(k) -40.2	(l) 22.4	(m) -41.9	(n) 23.0	(o) -44.1	(p) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-16.2	-14.8	-6.5	6.1	13.4	18.7	19.8	18.4	12.1	4.4	-5.2	-12.7
	DBStd	14.31	7.53	7.02	6.72	5.92	4.79	3.95	3.50	4.01	4.78	5.17	7.15	7.25
	HDD10.0	3556	812	693	511	142	22	1	0	1	32	183	455	704
	HDD18.3	5702	1070	927	769	366	161	42	26	49	194	431	705	963
	CDD10.0	1088	0	0	0	26	129	263	304	262	94	11	0	0
	CDD18.3	193	0	0	0	0	10	54	71	52	6	0	0	0
	CDH23.3	2564	0	0	0	18	199	718	777	686	163	3	0	0
CDH26.7	882	0	0	0	2	46	256	274	247	56	0	0	0	
Wind	WSAvg	4.1	4.2	4.4	4.7	4.3	4.3	3.8	3.6	3.5	3.6	3.9	4.3	4.6
Precipitation	PrecAvg	315	17	14	13	20	33	38	50	38	23	27	20	18
	PrecMax	443	53	41	34	66	83	119	163	178	66	70	54	56
	PrecMin	224	2	2	2	0	5	3	0	0	1	0	2	1
	PrecStd	56	11	8	8	14	18	25	40	34	16	19	12	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.3	11.2	26.0	31.0	34.1	34.9	34.4	32.1	22.9	10.2	1.7
		MCWB	0.5	0.4	6.5	13.3	16.0	17.9	18.3	17.8	16.6	10.9	7.3	0.7
	2%	DB	-0.9	-0.5	6.1	22.0	27.7	31.8	32.0	31.8	27.9	19.2	7.2	0.1
		MCWB	-1.4	-1.4	3.6	11.7	15.1	17.5	18.2	17.6	14.7	10.0	5.0	-0.4
	5%	DB	-3.0	-2.1	3.7	19.3	25.6	29.8	29.8	29.2	24.4	16.3	5.0	-1.2
		MCWB	-3.6	-2.7	2.1	10.4	14.0	17.0	17.9	17.1	13.5	9.0	3.6	-1.9
	10%	DB	-5.0	-4.0	2.0	16.3	23.3	27.7	27.6	27.1	21.5	13.1	3.0	-2.9
		MCWB	-5.5	-4.8	0.8	9.0	13.3	16.5	17.6	16.5	12.4	7.5	1.7	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.5	7.2	14.0	18.0	20.1	20.7	20.2	17.3	12.4	8.2	0.8
		MCDB	0.9	1.3	10.3	23.0	26.2	28.8	28.9	28.7	28.7	19.6	9.3	1.1
	2%	WB	-1.3	-1.4	3.8	12.5	16.2	19.1	19.7	19.0	15.5	10.8	5.4	-0.4
		MCDB	-0.9	-0.8	5.7	20.1	24.6	28.0	27.1	27.2	25.1	17.3	6.8	0.1
	5%	WB	-3.6	-2.9	2.1	11.1	15.1	18.2	19.0	18.1	14.2	9.4	3.4	-2.0
		MCDB	-3.1	-2.2	3.4	18.0	23.1	26.6	26.4	26.5	22.6	15.2	4.7	-1.5
	10%	WB	-5.6	-4.8	0.8	9.6	14.1	17.3	18.4	17.2	13.0	8.0	1.6	-3.5
		MCDB	-5.0	-4.1	1.7	15.7	21.9	25.2	25.4	25.5	20.6	12.6	2.8	-2.9
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.6	9.8	12.0	13.5	13.1	12.1	13.3	13.7	10.9	7.9	8.9
		MCDBR	7.7	9.3	9.4	17.2	17.0	16.6	16.0	17.2	19.0	16.9	8.9	8.4
		MCWBR	7.5	8.8	7.1	8.4	7.6	6.4	6.0	6.6	8.7	9.6	7.0	8.0
	5% WB	MCDBR	7.8	8.7	8.8	15.4	14.7	14.2	13.2	14.8	17.1	15.0	8.1	7.9
		MCWBR	7.6	8.5	6.9	8.1	7.4	6.3	5.6	6.4	8.3	9.0	6.7	7.7
Clear-Sky Solar Irradiance	taub	0.293	0.325	0.375	0.410	0.404	0.410	0.430	0.409	0.359	0.342	0.327	0.297	
	taud	2.056	1.998	1.970	2.133	2.219	2.267	2.214	2.253	2.346	2.301	2.146	2.048	
	Ebn at Noon	686	766	799	825	851	846	820	818	825	758	641	605	
	Edh at Noon	82	115	144	139	133	128	133	122	99	85	76	71	
All-Sky Solar Radiation	RadAvg	1.40	2.52	3.85	4.80	5.99	6.37	5.85	5.20	3.90	2.15	1.03	0.94	
	RadStd	0.13	0.10	0.37	0.41	0.51	0.50	0.41	0.37	0.27	0.29	0.11	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page